ClipSAT : Contest Math & Olympiad AMC 10/12, Kangaroo & Mawhiba

Unit 6.01: Olympiad Circle Geometry, Power of a Point & Radical Axes

Name: _____

Q1. The coordinate diagram shows the parabolic curve f_1x = x^2 - 16x + 80 in the domain of Circle Geometry, Power of a Point & Radical Axes. By completing the square f_1x = x - h^2 + k , find the sum k + f_110 .

[ ] A: 44
[ ] B: 36
[ ] C: 20
[ ] D: 34

Q2. A mathematical model in Circle Geometry, Power of a Point & Radical Axes is defined by the cubic function y_2x = 4x^3 - 8x^2 + 15x - 13 . Calculate the exact gradient of the tangent to the curve at the point where x = 3 .

[ ] A: 123
[ ] B: 60
[ ] C: 75
[ ] D: 35

Q3. In Circle Geometry, Power of a Point & Radical Axes, two displacement vectors are given by mathbf{u}_{3} = langle 8, -8 rangle and mathbf{v}_{3} = langle 9, 7 rangle . Compute the exact scalar value of mathbf{u}_{3} cdot mathbf{v}_{3} + |mathbf{u}_{3}|^2 .

[ ] A: 16
[ ] B: 128
[ ] C: 256
[ ] D: 144

Q4. The linear growth profile shown in the diagram for Circle Geometry, Power of a Point & Radical Axes follows an arithmetic progression with first term a_1 = 9 and common difference d = 6 . Find the exact sum S_{15} of the first 15 terms.

[ ] A: 765
[ ] B: 93
[ ] C: 855
[ ] D: 1395

Q5. In Circle Geometry, Power of a Point & Radical Axes, evaluate the exact definite integral I_5 = int_0^{2} 18x^2 + 32x,mathrm{d}x .

[ ] A: 136
[ ] B: 112
[ ] C: 144
[ ] D: 57

Q6. In Circle Geometry, Power of a Point & Radical Axes, consider the quadratic function f_6x = x^2 - 18x + 169 . Express f_6x in the completed-square form x - h^2 + k and compute the exact value of k + f_616 .

[ ] A: 234
[ ] B: 137
[ ] C: 225
[ ] D: 218

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Q7. The graph illustrates the cubic model y_7x = 4x^3 - 9x^2 + 33x - 20 studied in Circle Geometry, Power of a Point & Radical Axes. Find the instantaneous gradient frac{mathrm{d}y}{mathrm{d}x} of the tangent to the curve at x = 8 .

[ ] A: 801
[ ] B: 624
[ ] C: 111
[ ] D: 657

Q8. In Circle Geometry, Power of a Point & Radical Axes, two displacement vectors are given by mathbf{u}_{8} = langle 9, -7 rangle and mathbf{v}_{8} = langle 8, 5 rangle . Compute the exact scalar value of mathbf{u}_{8} cdot mathbf{v}_{8} + |mathbf{u}_{8}|^2 .

[ ] A: 167
[ ] B: 37
[ ] C: 130
[ ] D: 237